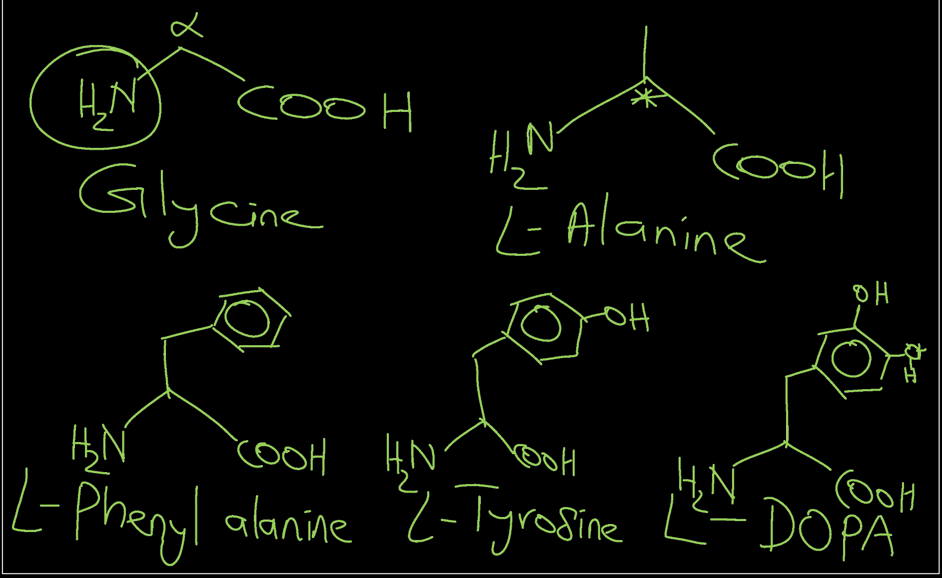
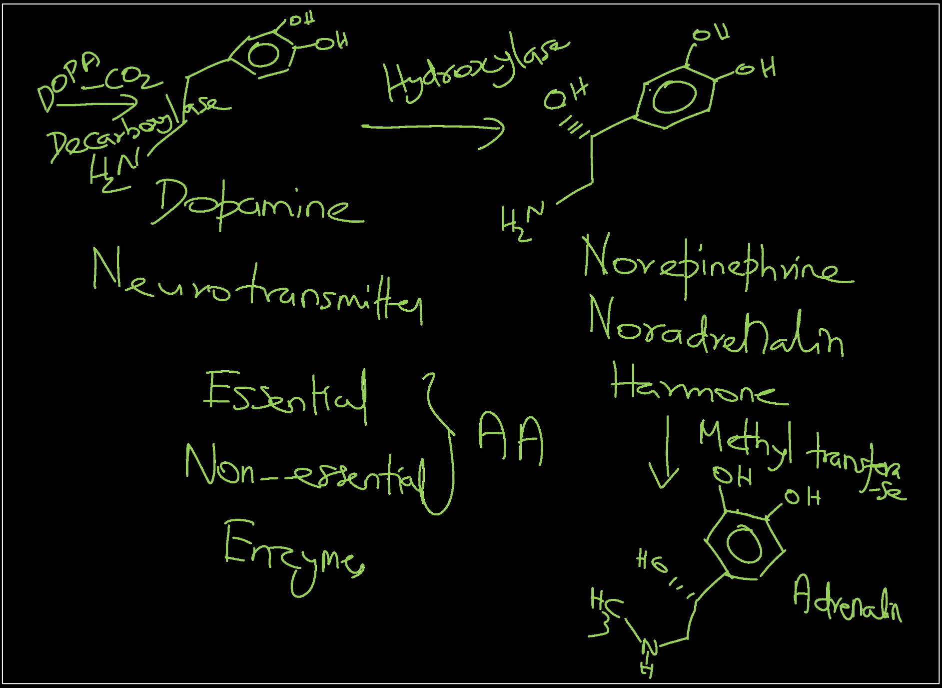
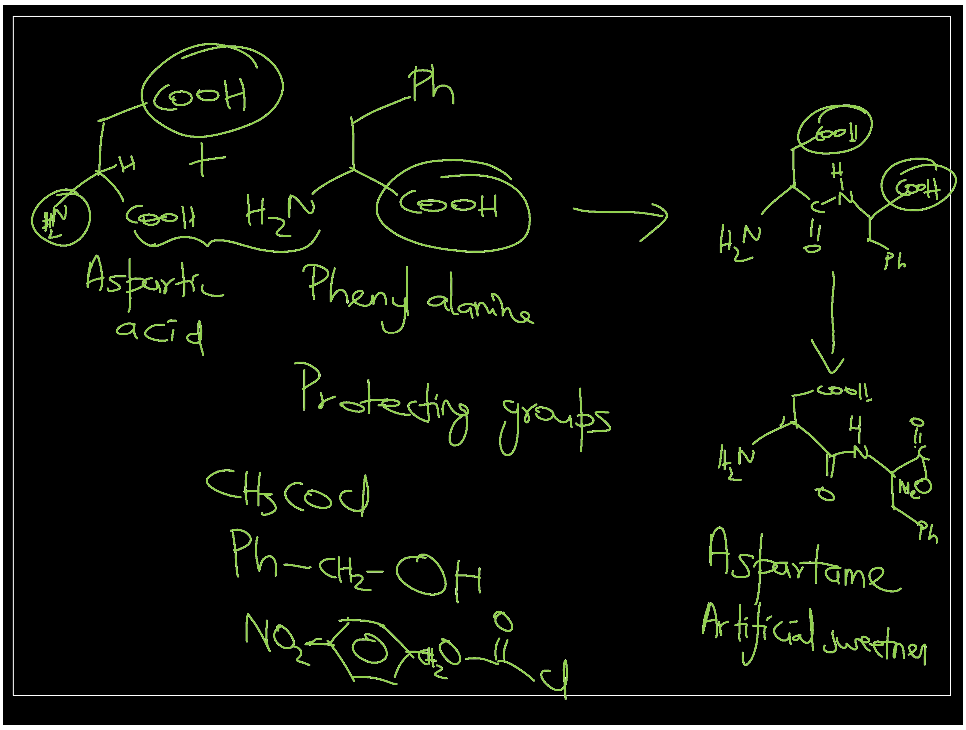
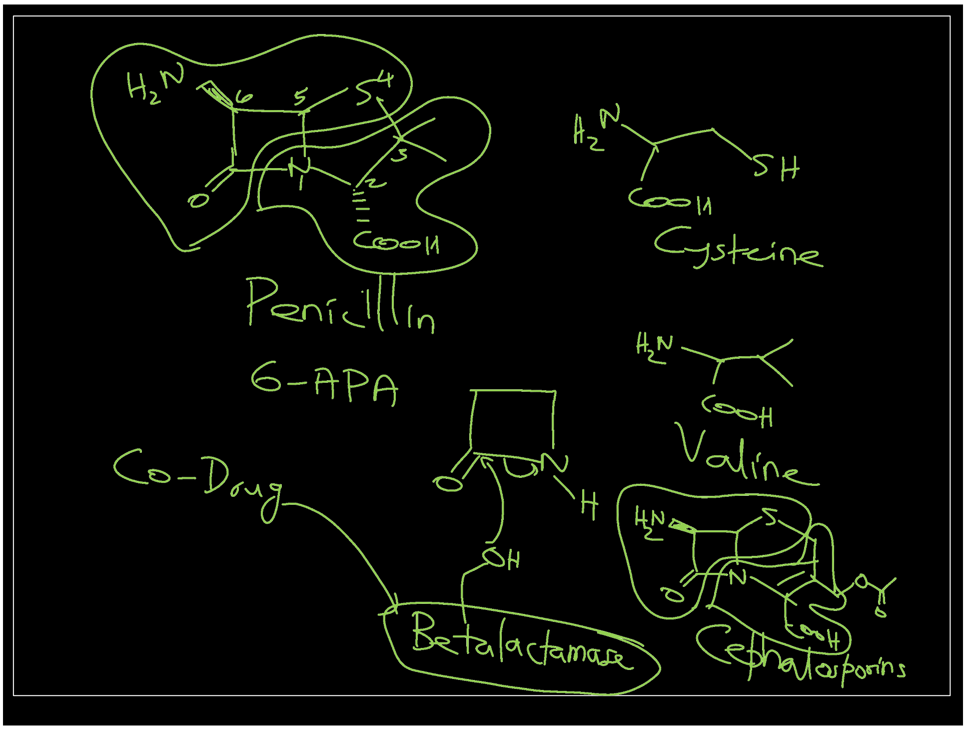
CHM102: Basic organic chemistry

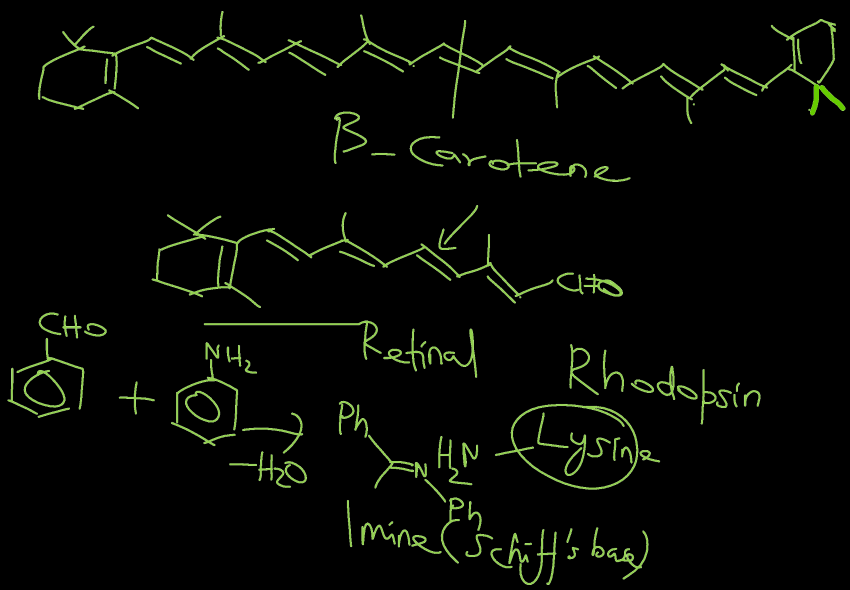
- A basic introduction to the **importance of organic molecules in life**.
- Nomenclature of organic compounds with diverse structure and functional groups.

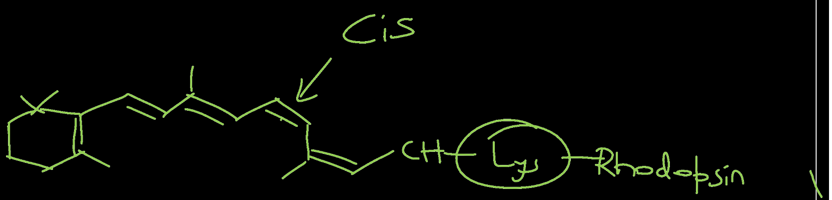





Aspartic acid
Phenyl alanine
Protecting groups
CH3COCl
Ph–CH2–OH
Aspartame
Artificial sweetner


Penicillin
6-APA
Co-Drug
Betalactamase
Cysteine
Valine
Cephalosporins

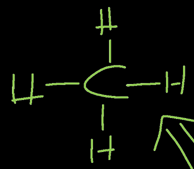

β- Carotene
CHO
NH2
Retinal
Rhodopsin
-H2O
Ph
H2N
Lysine
Ph
Imine (Schiff's base)

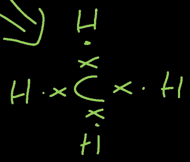

Cis
CH
Lys
Rhodopsin

CHM102: Basic organic chemistry

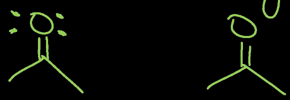
- Nomenclature of organic compounds with diverse structure and functional groups.

Lewis structure





Line drawing / Kekule structure



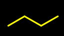
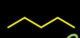
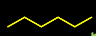
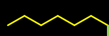
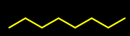
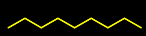
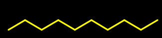
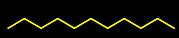
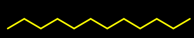
$CH_3\overset{O}{\overset{\|}{C}}CH_3$

$CH_3C(O)CH_3$

$CH_3C(=O)CH_3$

$CH_3CH_2CH_3$

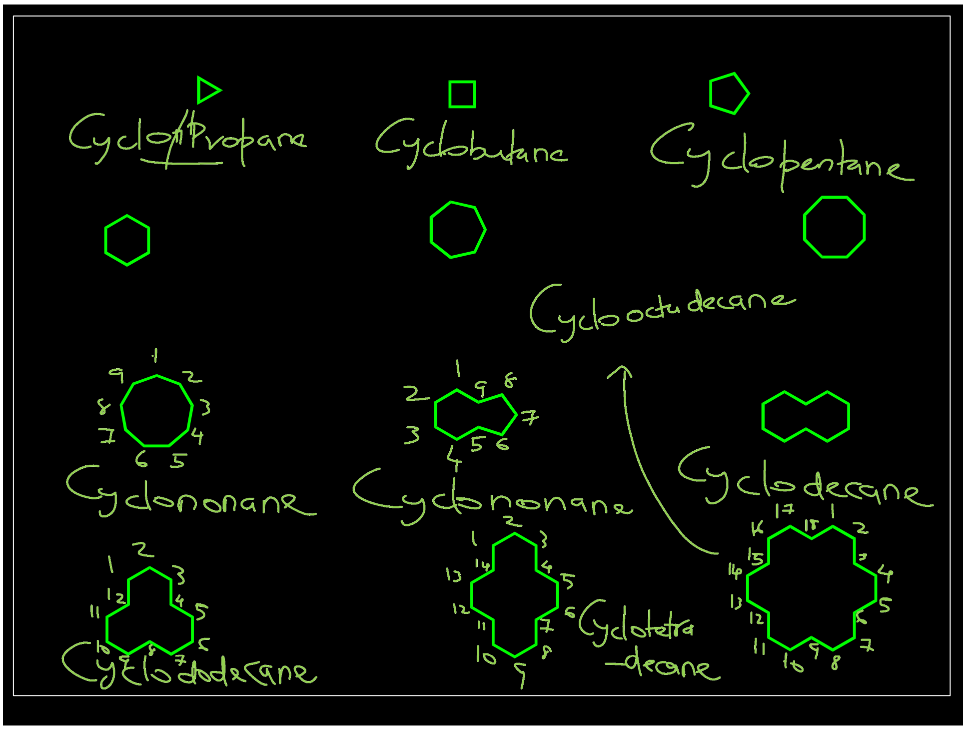
Condensed structure

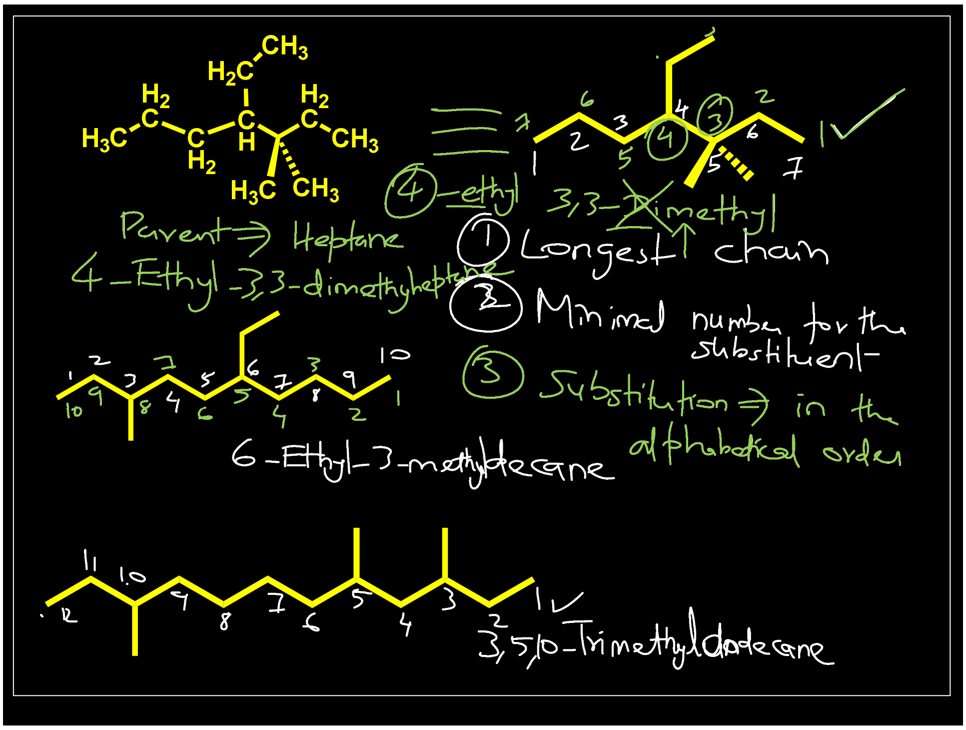
CH_4		Methane
CH_3CH_3		Ethane
$CH_3CH_2CH_3$		Propane
$CH_3CH_2CH_2CH_3$		n-Butane
$CH_3CH_2CH_2CH_2CH_3$		n-Pentane
$CH_3CH_2CH_2CH_2CH_2CH_3$		n-Hexane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$		n-Heptane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$		Octane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$		Nonane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$		Decane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$		Undecane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$		Dodecane (C_{12})

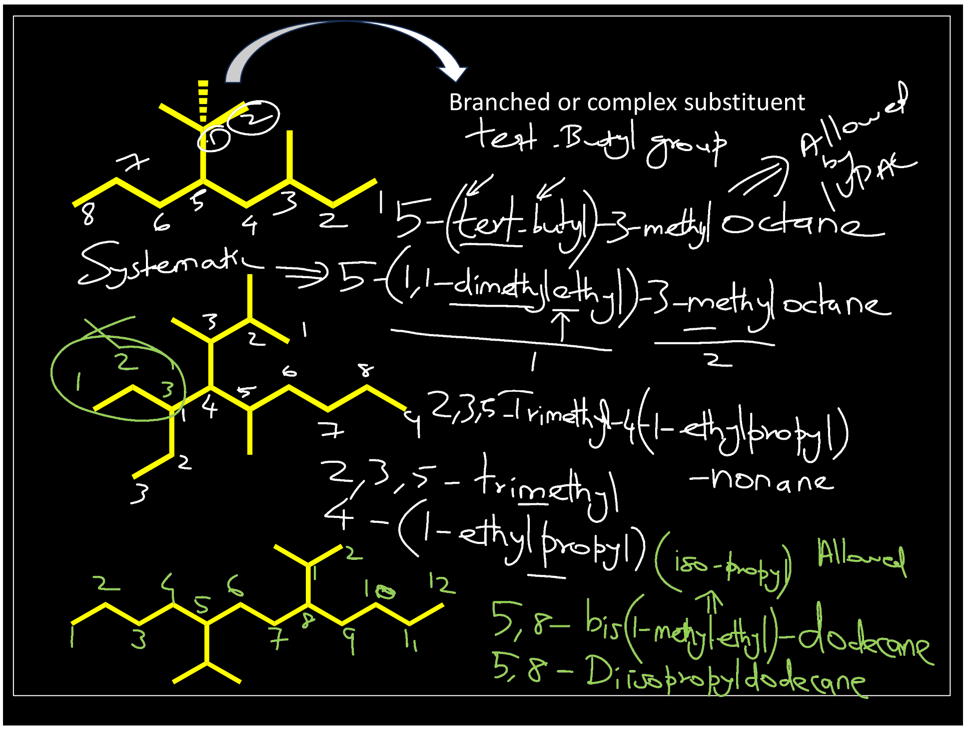
C_{13} Tridecane

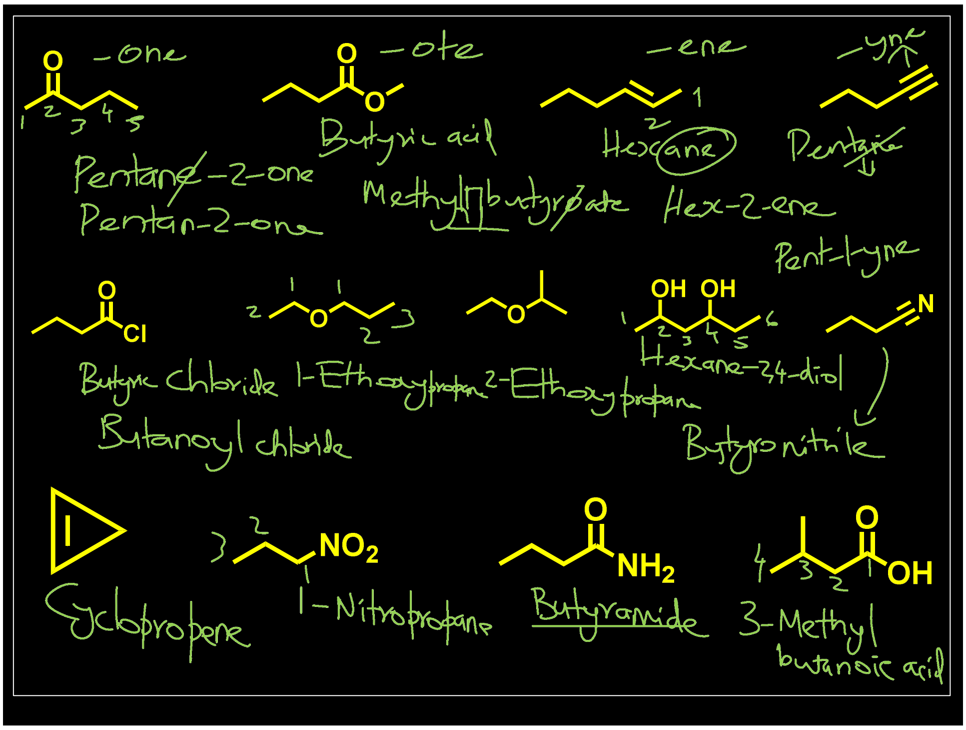
C_{14} Tetradecane

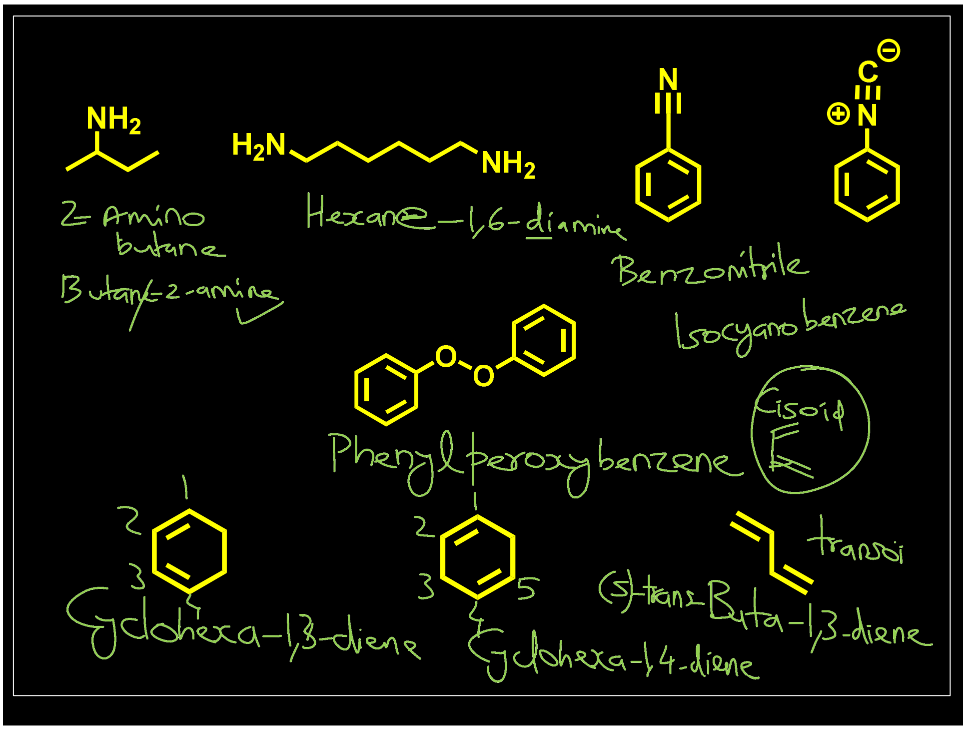
C_{20} Icosane

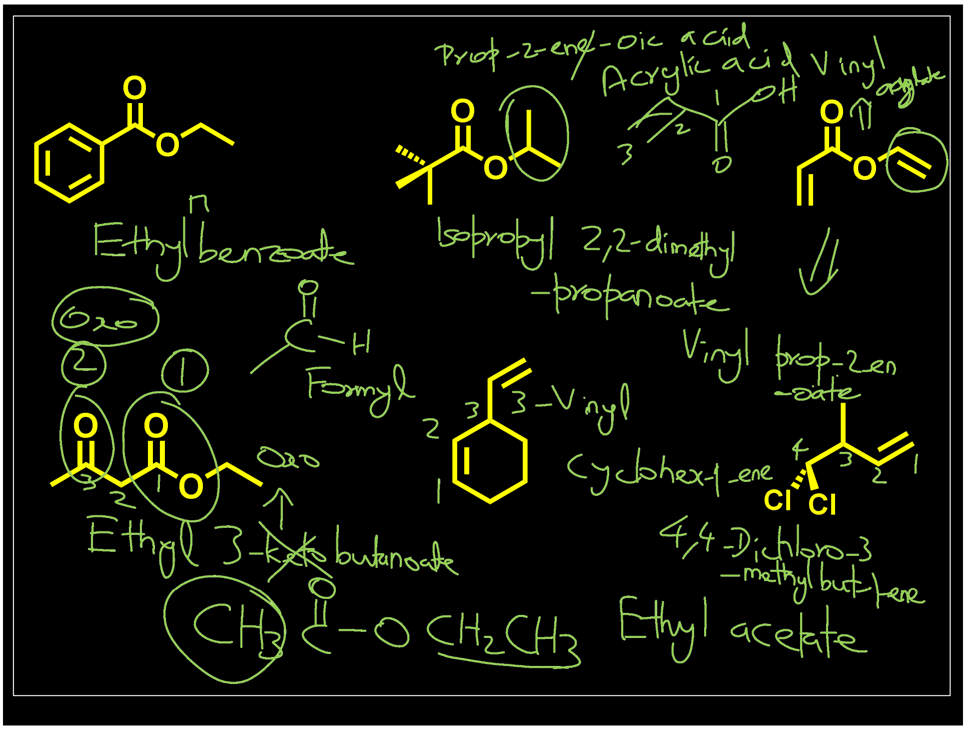

Cyclopropane
Cyclobutane
Cyclopentane
Cyclooctadecane
Cyclononane
Cyclononane
Cyclodecane
Cyclododecane
Cyclotetra
-decane


Parent ⇒ Heptane
4-Ethyl-3,3-dimethylheptane
4-ethyl
3,3-Dimethyl
1 Longest chain
2 Minimal number for the substituent
3 Substitution ⇒ in the alphabetical order
6-Ethyl-3-methyldecane
3,5,10-Trimethyldodecane


Branched or complex substituent
tert. Butyl group
Allowed by IUPAC
5-(tert-butyl)-3-methyl octane
Systematic ⇒ 5-(1,1-dimethylethyl)-3-methyl octane
2,3,5-Trimethyl-4-(1-ethylpropyl)
2,3,5-trimethyl
4-(1-ethyl propyl)
-nonane
(iso-propyl) Allowed
5,8-bis(1-methylethyl)-dodecane
5,8-Diisopropyldodecane


-one
-oate
-ene
-yne
Pentane-2-one
Pentan-2-one
Butyric acid
Methyl butyrate
Hexane
Hex-2-ene
Pentyne
Pent-1-yne
O
Cl
OH OH
N
Butyric Chloride
Butanoyl chloride
1-Ethoxypropane
2-Ethoxypropane
Hexane-2,4-diol
Butyronitrile
NO_2
NH_2
OH
Cyclopropene
1-Nitropropane
Butyramide
3-Methyl butanoic acid


NH2
2-Amino butane
Butane-2-amine
H2N NH2
Hexane-1,6-diamine
N
Benzonitrile
C
N
Isocyano benzene
O O
Phenyl peroxy benzene
Cisoid
transoid
(S)-trans Buta-1,3-diene
Cyclohexa-1,3-diene
Cyclohexa-1,4-diene


O
O
Ethyl benzoate
Prop-2-ene-oic acid
Acrylic acid
OH
Vinyl acrylate
Isopropyl 2,2-dimethyl -propanoate
Vinyl prop-2en-oate
Oxo
Formyl
3-Vinyl Cyclohex-1-ene
Cl Cl
4,4-Dichloro-3-methyl but-1-ene
Ethyl 3-~~Keto~~ butanoate
CH3 C-O CH2CH3
Ethyl acetate

Nomenclature – IUPAC rules

- Identify the longest carbon chain containing the functional group with the highest priority. This chain determines the *stem* or *parent name* of the compound.
- Identify the substituents attached to the long alkyl chain
- Numbering of the long alkyl chain begins at the end closest to the substituent(s)
- Alphabetic ordering of substituents
- Systematic names of complex substituents should be enclosed in parenthesis
- If complex substituent appears multiple times, prefixes (bis, tris, tetrakis, pentakis) should be used. (   )
- Change the ending of the parent alkane/alkene/alkyne to the *suffix* of the highest priority group, which gives the parent name of the compound (usually, drop the last letter "e" before adding the suffix, except for nitrile where the "e" is kept).
- Number the chain from the end closest to the highest functional group.
- The other groups are named as substituents by using the appropriate *prefixes*.
- Assign stereochemistry, E/Z or R/S, as necessary

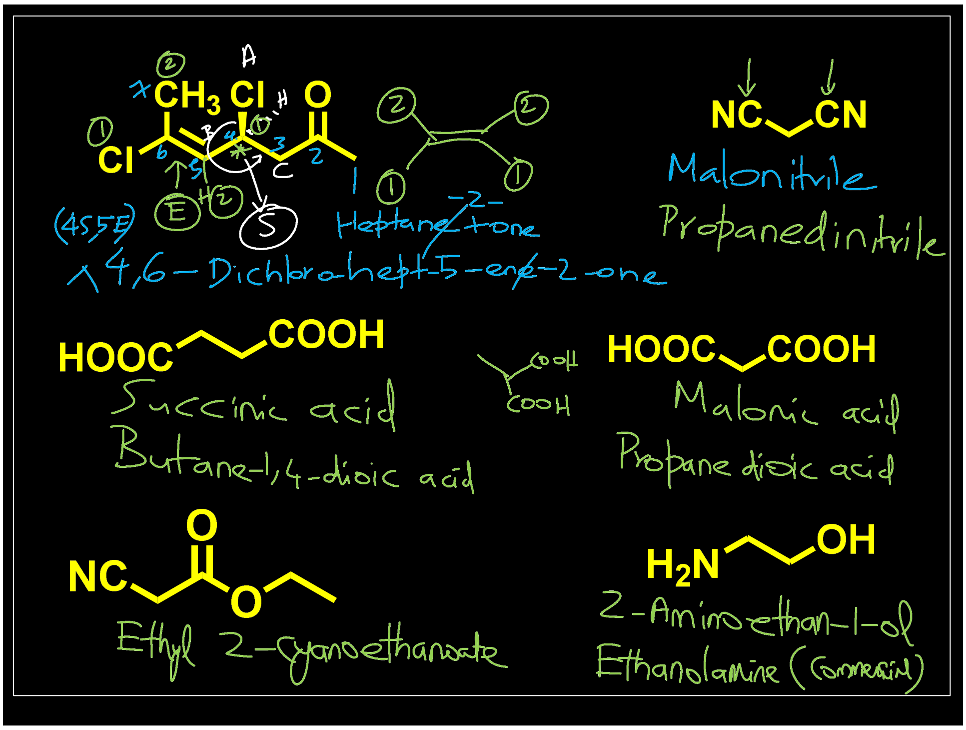
Table 3: Seniority order for characteristic groups.

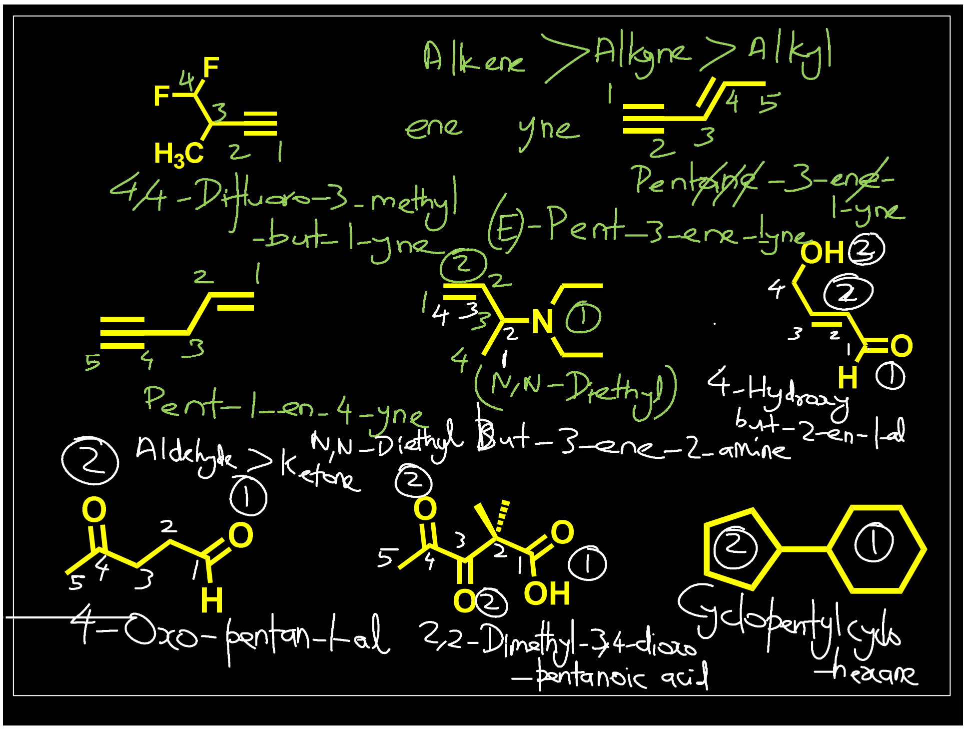
Priority decreases

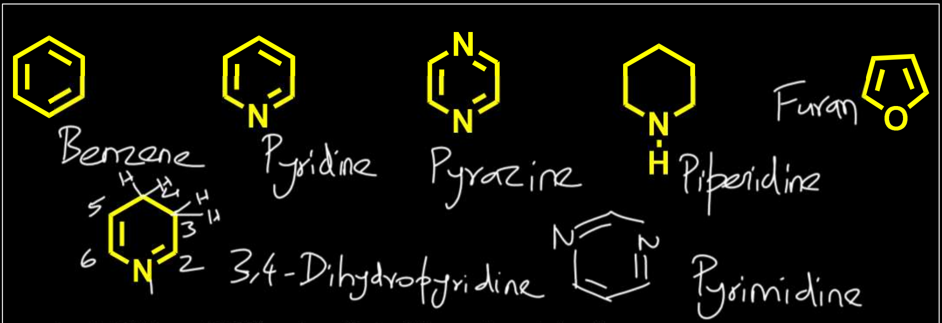
Class	Formula*	Suffix	Prefix
Carboxylates	$-COO^-$	carboxylate	carboxylato
	$-(C)OO^-$	oate	
Carboxylic acids	−COOH	carboxylic acid	carboxy
	−(C)OOH	oic acid	
Esters	−COOR	(R) …carboxylate**	(R)oxycarbonyl
	−(C)OOR	(R) …oate**	
Acid halides	−COX	carbonyl halide	halocarbonyl
	−(C)OX	oyl halide	
Amides	$-CONH_2$	carboxamide	carbamoyl
	$-(C)ONH_2$	amide	
Nitriles	−C≡N	carbonitrile	cyano
	−(C)≡N	nitrile	
Aldehydes	−CHO	carbaldehyde	formyl
	−(C)HO	al	oxo
Ketones	=O	one	oxo
Alcohols	−OH	ol	hydroxy
Thiols	−SH	thiol	sulfanyl***
Amines	$-NH_2$	amine	amino
Imines	=NH	imine	imino

* Here −(C) indicates that the carbon atom is implied by the parent name.
** Here (R) means that the group R is expressed as a separate prefixed word.
*** Note: 'mercapto' is no longer acceptable (but is still used by CAS).

Pure Appl. Chem. **2020**; *92*(3): 527–539


4,6-Dichlorohept-5-ene-2-one
Heptane-2-one
NC CN
Malonitrile
Propanedinitrile
HOOC COOH
Succinic acid
Butane-1,4-dioic acid
Malonic acid
Propane dioic acid
NC O
Ethyl 2-cyanoethanoate
H2N OH
2-Aminoethan-1-ol
Ethanolamine (commercial)


Alkene > Alkyne > Alkyl
ene yne
4,4-Difluoro-3-methyl-but-1-yne
(E)-Pent-3-ene-1-yne
Pent-1-en-4-yne
N,N-Diethyl But-3-ene-2-amine
4-Hydroxy but-2-en-1-al
Aldehyde > Ketone
4-Oxo-pentan-1-al
2,2-Dimethyl-3,4-dioxo-pentanoic acid
Cyclopentylcyclohexane



Hantzsch-Widman (H-W) system (3- to 10-membered rings)

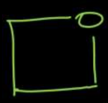
O	oxa	S	thia	N	aza	P	phospha
As	arsa	Si	sila	Sn	stanna	B	bora

Ring size	Unsaturated	Saturated
3	irene	irane
4	ete	etane
5	ole	olane
6	ine/ine/inine*	ane/inane/inane*
7	epine	epane

Hantzsch-Widman (H-W) system (3- to 10-membered rings)

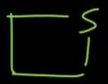
O	oxa	S	thia	N	aza	P	phospha
As	arsa	Si	sila	Sn	stanna	B	bora

Ring size	Unsaturated	Saturated
3	irene	irane
4	ete	etane
5	ole	olane
6	ine/ine/inine*	ane/inane/inane*
7	epine	epane



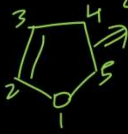
Ox~~a~~+etane

Oxetane



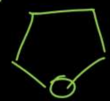
Thi~~a~~+etane

Thietane



4,5-Dihydro oxole

Ox~~a~~+ole



THF

Tetrahydrofuran

Ox~~a~~+olane

Oxolane

Oxole

Hantzsch-Widman (H-W) system (3- to 10-membered rings)

O	oxa	S	thia	N	aza	P	phospha
As	arsa	Si	sila	Sn	stanna	B	bora

Ring size	Unsaturated	Saturated
3	irene	irane
4	ete	etane
5	ole	olane
6	ine/ine/inine*	ane/inane/inane*
7	epine	epane